

# Hasan Zohirul Islam

Graduate Researcher – Machine Learning & Multimodal Learning

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## RESEARCH INTERESTS

**Multimodal Learning & Fusion:** Cross-modal fusion strategies, modality contribution analysis, and information-flow-guided selection of optimal fusion points in heterogeneous neural networks.

**Biomedical & Bioacoustic Signal Processing:** Deep learning for physiological signal analysis and environmental audio recognition (bioacoustic species identification).

**Image Forensics & Integrity Verification:** Detection and localization of image manipulations, with focus on copy-move forgery detection and pixel-level segmentation of tampered regions.

**Geometric Deep Learning & CNN Robustness:** Equivariant and rotation-invariant convolutional architectures; understanding and improving CNN robustness to geometric transformations.

## EDUCATION

**East West University, Bangladesh**

*Jan 2026 – Present*

M.Sc. in Computer Science & Engineering

CGPA: 3.89 / 4.00

*Major: Data Science and AI track; Expected Completion: December 2026*

**M.Sc. Thesis:** *Fused Accuracy Does Not Imply Both Branches Learned: Silent Branch Collapse in Dual-View Biomedical Audio*

*What we did:* Showed that a published dual-view (spectrogram + waveform) heart-sound classifier’s spectrogram branch – 72% of its parameters – learns nothing despite strong fused accuracy, using branch ablation, an untrained-control linear probe, and a standalone-training criterion. Traced the failure to gradient starvation rather than an information deficit, under a pre-registered protocol with a sealed test set.

**Bangladesh University of Engineering and Technology (BUET)**

*2016 – 2021*

B.Sc. in Electrical and Electronic Engineering

## TEST SCORES

**GRE:** 322/340 (Quantitative: 168, Verbal: 154)

**TOEFL iBT:** 97/120 (R23, L26, S23, W25)

## RESEARCH EXPERIENCE

**M.Sc. Thesis Research, East West University**

*Jan 2026 – Present*

*Graduate Researcher*

Dhaka, Bangladesh

- Designed a pre-registered protocol – branch ablation, an untrained-control linear probe, a standalone-training criterion, and a sealed test set – to diagnose whether a trained branch of a dual-view model had actually learned anything.
- Ran the diagnostic across three biomedical/bioacoustic corpora to establish the boundary conditions under which the failure does and does not occur.

**Analytics for Cyber Defense (ACyD) Lab, Florida International University**

*Mar – Jun 2023*

*Graduate Research Assistant*

Miami, USA

- Researched **adversarial attacks on distributed deep learning models** in edge devices; analyzed vulnerabilities and potential defense mechanisms.

## PUBLICATIONS

- Islam, H. Z. et al. **Fused Accuracy Does Not Imply Both Branches Learned: Silent Branch Collapse in Dual-View Biomedical Audio.** *Manuscript in preparation, to be submitted.*
- Islam, H. Z. et al. (2024). **Muzzle-Based Cattle Identification System Using AI.** *arXiv:2407.06096.*

- Sarker, M. S. A., Islam, H. Z. et al. (2022). **Seaweed Use for Reducing Enteric Methane in Cattle**. *Veterinary Research Notes*, 2(11), 78–85. doi:10.5455/vrn.2022.b18.

## HONORS & AWARDS

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- **2nd Runner-Up – IEEE MIPR AVA Challenge (2024)**: Developed video-analysis models for imminent accident prediction from large-scale dashcam data; contributed to autonomous driving safety research.
- **Champion – Datathon 3.0, Robi Axiata Ltd. (2024)**: Outperformed 1000+ teams in Bangladesh’s largest data science competition with advanced ML solutions for telecom challenges.
- **1st Runner-Up – Telenor Global Hackathon (2025)**: AI call-center solution featuring (i) real-time agent evaluation via ASR transcription and tone/empathy analysis, and (ii) live sign-language recognition with AI-avatar responses for deaf customers.
- **Champion – MATLAB Contest, CUET (2020)**.
- Divisional Math Olympiad **Champion** (2011) and **1st Runner-Up** (2012).

## TECHNICAL SKILLS

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- **Machine Learning & Deep Learning**: PyTorch, TensorFlow, Scikit-learn; CNNs, Transformers, Multimodal Fusion, Time-Series & Signal Modeling
- **Computer Vision & Audio**: OpenCV, Image Segmentation, Object Detection, OCR, Spectrogram-based Audio Classification
- **Generative AI & LLMs**: RAG Pipelines, Agentic Systems, Text-to-SQL, Vector Databases
- **Programming**: Python (primary), C++, SQL, MATLAB
- **MLOps & Deployment**: Docker, Kubeflow, MLflow, FastAPI, CI/CD, Azure AI Foundry

## PROFESSIONAL EXPERIENCE

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**Grameenphone Ltd. (Telenor Group)** *Apr 2025 – Present*  
*AI Engineer, Senior Executive* Dhaka, Bangladesh

- **Network Operations AI Assistant (Agentic RAG + Text-to-SQL)**: Designed and deployed a dual-mode agentic AI assistant for field technicians across the national tower network, combining a retrieval-augmented generation (RAG) knowledge base with a Text-to-SQL module for natural-language queries over live tower and alarm data. Deployed on Azure AI Foundry.
- **OneAI – Employee Policy Assistant**: Built a policy bot for the company’s employee apps that answers natural-language policy queries, replacing manual lookup. Deployed on Azure.
- **Customer Churn Prediction**: Developed and productionized churn-risk prediction models feeding targeted retention workflows.
- **ERS Out-of-Stock Prediction**: Built a real-time stock-depletion forecasting model running on an on-premise Kubernetes cluster to enable proactive restocking.
- **Recommendation Engine (Ongoing)**: Building a personalized recommendation system predicting the best-fit pack for each customer from usage behaviour.

**Advanced Chemical Industries Ltd. (ACI)** *Sept 2023 – Mar 2025*  
*Machine Learning Engineer* Dhaka, Bangladesh

- Developed a real-time computer vision system for automated vehicle identification at factory gates, deployed across 10 factories, including Bangla license plate recognition.
- Built a face recognition platform supporting automated attendance and customer auto-identification, deployed across multiple sites.

**Adorsho Pranisheba** *Jul 2021 – Nov 2022*  
*Machine Learning Engineer* Dhaka, Bangladesh

- **Cattle Muzzle Recognition (Biometric Identification)**: Curated a dataset of 50,000 images spanning ~10,000 cattle and fine-tuned FaceNet with triplet-loss metric learning for biometric identification.
- **Cattle Health & Activity Monitoring System**: Built a multimodal system combining embedded IoT sensors and computer vision to detect behavioural anomalies in real time.

## LEADERSHIP & SERVICE

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- **Associate General Secretary** – Satyen Bose Science Club, BUET.
- **Academic Team Member** – Bangladesh Mathematical Olympiad Committee.